

Comp 2310-lab 3

1- with these hypotheses

$$P1: \sim \exists x (P(x) \wedge B(x))$$

$$P2: \sim \exists x (S(x) \wedge C(x))$$

$$P3: \forall x (\sim B(x) \Rightarrow C(x))$$

Prove that $\sim \exists x (P(x) \wedge S(x))$

2- Show that

- A person in the hospital has covid

- Every person in the hospital has been healed

would imply that "A person with covid has been healed"

Use $H(x)$: x is in hospital

$C(x)$: x has covid

$L(x)$: x has been healed

3- Prove that $A \neq B$ if and only if

$$\exists x (x \in A \wedge x \notin B) \vee \exists x (x \notin A \wedge x \in B).$$

4- Using set-builder notation and definitions of $=$ and $\subseteq$ prove the following statements for sets A, B, C :

(a) $A \subseteq A$

(b) if $A \subseteq B$ and $B \subseteq A$ then $A = B$

(c) if $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$

(d) $A - (A - B) = A \cap B$

(e) $(A \cap B) \times C = (A \times C) \cap (B \times C)$